PROBABILITY AND STATISTICS

MID-TERM EXAM SYLLABUS & COMPLETE CLASS NOTES

SYLLABUS OUTLINE

CHAPTER 1	CHAPTER 2	CHAPTER 3
• Statistics • Branches of Statistics • Descriptive Statistics • Inferential Statistics • Population • Sample • Parameter • Statistic • Statistical Data • Primary Data • Secondary Data • Variable • Continuous Variable • Qualitative Variable • Quantitative Variable • Discrete Variable • Continuous Variable	• Presentation of Data • Types/Methods of Presentation of Data • Classification of Data • Types of Classification of Data • Tabulation • Charts • Types of Charts • Graphs • Methods for Collecting Primary Data • Methods for Collecting Secondary Data • Main Parts of a Table	• Average • Types of Average • Mean • Mode • Median • Stem-and-Leaf • Stem • Leaf • Frequency Distribution • Grouped Data • Ungrouped Data • Class Limits • Class Boundaries • Class Interval • Class Frequency • Central Tendency • Statistical Average • Arithmetic Mean • Properties of Arithmetic Mean • Properties of Mode • Properties of Median

CHAPTER 1

Overview Flowcharts (From Notes)

Statistics Pillars: Statistics → **C** (Collection) | **P** (Presentation) | **A** (Analyzation) | **I** (Interpretation)

Core Hierarchy: Population → Sample → Parameter → Statistic

Variable Breakdown: Variable → Qualitative & Quantitative. Quantitative Variable → Discrete variable (Countable) & Continuous variable (Uncountable).

Statistics

It is a branch of science which deals with the collection, presentation, analyzation and interpretation of numerical data.

Branches of Statistics

- **Descriptive Statistics**
- **Inferential Statistics**

Descriptive Statistics

It is the branch of statistics which deals with the collection, organization, analysis, and interpretation of numerical data and also any numerical measures.

Inferential Statistics

It is the branch of statistics which deals with making inferences about a population on the basis of sample data, and also estimates the population parameter.

Population

The group under study or discussion is called population. It is always denoted by **N**. It includes a group of living and non-living things.

Sample

A small representative part of a population is called a sample. It is denoted by **n**.

Parameter

A numerical value which is calculated from a population is called a parameter. *(Note: How to represent a parameter? It is calculated directly from population data.)*

Statistic

A numerical value which is calculated from a sample is called a statistic.

Statistical Data

Raw facts and figures. A collection of facts and figures in numerical form obtained either by counting or measuring.

- **Primary Data**
- **Secondary Data**

Primary Data

The most original and fresh data. Such type of data that has not undergone any statistical treatment is called primary data.

Secondary Data

It is the second-hand information. The data which has undergone statistical treatment is called secondary data.

Variable (and Constant)

Variable: A quantity which does not remain fixed. It is denoted by capital letters (e.g., X, Y) and its values are denoted by small letters (e.g., x, y).

Constant: A quantity which remains fixed.

Continuous Variable

An uncountable variable that can take any numerical value within a range (obtained by measurement).

Qualitative Variable (Including Nominal & Ordinal Data)

Qualitative Variable: A variable which can change in quality from one object to another.

- **Nominal Data:** The values are just names or labels; you cannot meaningfully say one category is higher or lower than another. *Examples:* Religion, Colour, Blood group.
- **Ordinal Data:** The values can be defined and ranked from lower to higher, but the exact difference between categories is not known. *Examples:* Rankings, Poor / Good / Excellent, Positions, Educational status.

Quantitative Variable

A variable which can change in quantity from one object to another.

Discrete Variable

Countable variable (takes specific, whole-number countable values).

Continuous Variable

Uncountable variable (takes continuous, measurable values across an interval).

CHAPTER 2

Presentation of Data

The collected data organized and summarized in a form that is easy to understand is called presentation of data.

Types/Methods of Presentation of Data

- **Classification**
- **Tabulation**
- **Charts**
- **Graphs**

Classification of Data

The process of arranging data into homogeneous classes or groups according to their similarities present in the data is called classification.

Types of Classification of Data

- **One-way Classification:** When the data is classified by one characteristic, it is called one-way classification.
- **Two-way Classification:** When the data is classified by two characteristics, it is called two-way classification.

Tabulation

The process of arranging data into rows and columns is called tabulation.

Charts

A chart is the device used to represent simple statistical data in a simple, clear, and effective manner.

Types of Charts

- **Simple bar chart**
- **Multiple bar chart**
- **Component bar chart**
- **Percentage component bar chart**
- **Pie chart (Pie-bar chart)**

Graphs

It represents the data in the form of lines and curves to show the fluctuations, trends, and other features of the data.

Types of Graphs:

- **Histogram**
- **Frequency polygon**
- **Frequency curve**
- **Cumulative frequency polygon (Ogive)**

Methods for Collecting Primary Data

- **Direct Investigation**
- **Indirect Investigation**
- **Registration method**
- **Questionnaire method**
- **Through telephone**
- **Through local sources**

Methods for Collecting Secondary Data

- **Official sources**
- **Semi-official sources**
- **Private sources**
- **Research Organizations**

Main Parts of a Table

- **Title**
- **Column caption**
- **Row caption (Stub)**
- **Box Head**
- **Body**
- **Footnote**
- **Source note**

CHAPTER 3

Average

An average is a single value which represents all values of the data; it is called an average.

Types of Average

- **Arithmetic Mean (A.M)** — denoted by $\bar{X}$
- **Geometric Mean (G.M)**
- **Harmonic Mean (H.M)**
- **Mode** — denoted by $\hat{X}$ (most repeated value)
- **Median** — denoted by $\tilde{X}$

Mean

The Arithmetic Mean is the sum of all elements divided by the number of elements.

> **A.M** = (Sum of all elements) / (Number of elements)
> Where $\bar{X}$ = mean, X = data, n = number of elements, f = frequency.
> **Ungrouped Data Formula:** $\bar{X} = (\sum X) / n$
> **Grouped Data Formula:** $\bar{X} = (\sum fx) / (\sum f)$

Mode

Mode ($\hat{X}$) is the most repeated value in the data.

> **Ungrouped Data:** $\hat{X}$ = Most repeated observation in the dataset.
> **Grouped Data Formula:** $\hat{X} = l + [\ (f_m - f_1) / ((f_m - f_1) + (f_m - f_2))\ ] \times h$
> Where l = lower class boundary of modal class, f_m = frequency of modal class (highest frequency), f_1 = frequency of class preceding the modal class, f_2 = frequency of class following the modal class, and h = class interval.

Median

Median ($\tilde{X}$) is the middlemost value of the arranged data (in ascending or descending order) which divides the total number of values into two equal parts.

> **Ungrouped Data:** Arrange values in ascending order and locate the middle value (for odd n, the $((n + 1) / 2)^{\text{th}}$ term; for even n, the average of the two middle terms).
> **Grouped Data Formula:** $\tilde{X} = l + (h / f) \times [\ (n / 2) - C\ ]$
> Where l = lower class boundary of median class, h = class interval size, f = frequency of median class, $n = \sum f$, and C = cumulative frequency of the class preceding the median class.

Stem-and-Leaf

We represent numerical data into stem and leaf. The first digit is the stem, and the second (and other trailing digit) is the leaf. A stem-and-leaf plot is a graphical tool used to organize and display numerical data.

- **Example 1 (Two-digit number):** In **23** → **2** is the *stem* and **3** is the *leaf*.
- **Example 2 (Three-digit number):** In **102** → **10** is the *stem* and **2** is the *leaf*.

Stem

The left part which consists of the first one or more digits.

Leaf

The right part which contains the last digit.

Solved Practical Questions: Stem-and-Leaf Plot

Solved Question 1 (Two-Digit Data): Construct a Stem-and-Leaf plot for the following exam scores of 12 students: **12, 15, 18, 21, 23, 23, 27, 30, 34, 38, 42, 45**.

Stem (Tens Digit)	Leaf (Units Digit)
1	2 5 8
2	1 3 3 7
3	0 4 8
4	2 5

Key: Stem 2 | Leaf 3 = 23.

Solved Question 2 (Three-Digit Data): Construct a Stem-and-Leaf display for the following measurements: **102, 105, 109, 114, 117, 120, 126, 128**.

Stem (First Two Digits)	Leaf (Last Digit)
10	2 5 9
11	4 7
12	0 6 8

Key: Stem 10 | Leaf 2 = 102.

Frequency Distribution

When the data is distributed into classes or groups with a number of observations falling in each group, it is called a frequency distribution.

Grouped Data

When the data is distributed and presented into a frequency distribution, it is called grouped data.

Ungrouped Data

Same definition as primary data: raw observations that have not been organized into classes or frequency distributions.

Class Limits

Each class is described by two numbers. The smaller number is the lower class limit, and the higher number is the upper (higher) class limit. These values are called class limits.

Class Boundaries

The true class limits are known as class boundaries. The smaller limit is the lower class boundary, and the larger limit is called the upper class boundary.

Class Interval (Class / Interval / Size / Gap)

The difference between the lower class boundary and upper class boundary is called the Class Interval. It is denoted by **h** or **c**.

Class Frequency

The number of observations falling in each class is called class frequency. It is denoted by **f**.

Central Tendency

The extent of observations to cluster around the central value is called central tendency.

Statistical Average

A statistical average is a single numerical summary value calculated from a dataset that represents the center of the distribution.

Arithmetic Mean

The sum of all values divided by the total number of values: $\bar{X} = (\sum x) / n$ (for ungrouped data) and $\bar{X} = (\sum f x) / (\sum f)$ (for grouped data).

Properties of Arithmetic Mean

- The mean of a constant is the constant itself ($c = c$).
- The sum of deviations of the values from their mean is always equal to zero: $\sum (X - \bar{X}) = 0$.
- The sum of squared deviations from the mean is minimum (least): $\sum (X - \bar{X})^2 = minimum$.

Properties of Mode

- Mode is the most repeated value and is very easy to locate by inspection in ungrouped data.
- It is not affected by extreme values (outliers).
- It can be determined for open-ended frequency distributions and qualitative data.
- A dataset may have one mode, more than one mode, or no mode at all.

Properties of Median

- Median is a positional average that divides an ordered dataset into two equal halves (50% below and 50% above).
- It is unique for every dataset (there is always only one median).
- It is not affected by extreme values (outliers) and can be computed for open-ended distributions.
- The sum of absolute deviations from the median is minimum: $\sum |X - \tilde{X}| = minimum$.

SOLVED PRACTICAL QUESTIONS (MEAN, MEDIAN & MODE — UNGROUPED & GROUPED DATA)

Practical Question 1: Arithmetic Mean / Mean (From Notes)

Given Dataset: $A = \{10, 12, 13, 15, 14, 17, 18\}$ (Total number of values $n = 7$)

a) Ungrouped Data Solution:

> Formula: $\bar{X} = (\sum x) / n$
> $\bar{X} = (10 + 12 + 13 + 15 + 14 + 17 + 18) / 7 = 99 / 7 =$ ***14.14***

b) Grouped Data Solution:

First, construct the frequency distribution from the raw data:

- **Number of classes:** $k = 1 + 3.3 \log(7) = 1 + 3.3(0.845) = 3.79 \approx 4$ (or 5 classes covering up to 18)
- **Class interval (h):** $h = Range / (Number\ of\ classes) = (18 - 10) / 4 = 8 / 4 = 2$

Classes	Class Boundaries (C.B)	Tally	Frequency (f)	Cumulative Frequency ($c.f$)	Midpoint (x)	fx
10 – 11	9.5 – 11.5	|	1	1	10.5	10.5
12 – 13	11.5 – 13.5	||	2	3	12.5	25.0
14 – 15	13.5 – 15.5	||	2	5	14.5	29.0
16 – 17	15.5 – 17.5	|	1	6	16.5	16.5
18 – 19	17.5 – 19.5	|	1	7	18.5	18.5
Total			**∑ f = 7**	—	—	**∑ f x = 99.5**

> Formula: $\bar{X} = (\sum fx) / (\sum f)$
> $\bar{X} = 99.5 / 7 =$ ***14.21***

Practical Question 2: Median (From Notes)

a) Ungrouped Data Solution:

Arrange the given data in ascending order ($n = 7$):

Ordered Data: *10, 12, 13,* ***14****, 15, 17, 18*
Middle Position = $(n + 1) / 2 = (7 + 1) / 2 = 4^{th}$ value
Median ($\tilde{X}$) = 14

b) Grouped Data Solution (Using the Frequency Table Above):

Formula: $\tilde{X} = l + (h / f) \times [\ (n / 2) - C\ ]$
1. Locate Median Class: $n / 2 = 7 / 2 = 3.5$. Looking at the *c.f* column, *3.5* falls in cumulative frequency **5**, which corresponds to the class **14 – 15** (Class Boundaries: **13.5 – 15.5**).
2. Identify values: Lower boundary $l = 13.5$, class interval $h = 2$, frequency of median class $f = 2$, preceding cumulative frequency $C = 3$.
3. Substitute into formula:
$\tilde{X} = 13.5 + (2 / 2) \times (3.5 - 3)$
$\tilde{X} = 13.5 + 1 \times (0.5) = 13.5 + 0.5 =$ ***14***

Practical Question 3: Mode (Ungrouped & Grouped Data)

a) Ungrouped Data Solution:

Consider the ungrouped dataset: $B = \{10, 12, 13, 14, 14, 15, 17, 18\}$

Formula: *Mode ($\hat{X}$) = Most repeated value in the dataset*
Here, **14** occurs 2 times while all other numbers occur only once.
Mode ($\hat{X}$) = 14
(Note: In the 7-value set {10, 12, 13, 15, 14, 17, 18}, every number appears once, so it has no mode.)

b) Grouped Data Solution (Using Your Grouped Frequency Table):

Formula: $\hat{X} = l + [\ (f_m - f_1) / ((f_m - f_1) + (f_m - f_2))\ ] \times h$
1. Locate Modal Class: The highest frequency is $f_m = 2$. Taking the class **12 – 13** (Class Boundaries: **11.5 – 13.5**):
2. Identify values: Lower boundary $l = 11.5$, modal frequency $f_m = 2$, preceding frequency $f_1 = 1$ (from class 10–11), following frequency $f_2 = 2$ (from class 14–15), class interval $h = 2$.
3. Substitute into formula:
$\hat{X} = 11.5 + [\ (2 - 1) / ((2 - 1) + (2 - 2))\ ] \times 2$
$\hat{X} = 11.5 + [\ 1 / (1 + 0)\ ] \times 2 = 11.5 + 2 =$ ***13.5***

IMPORTANT POINT

IMPORTANT FOR MID-TERM EXAM

- **Numerical Questions:** Stem-and-Leaf, Mean, Mode and Median.
- **All remaining topics:** Theoretical questions.